Apoptosis

LG3

Lecture Outline:

- Definition of Apoptosis
- Causes of Apoptosis
- Morphology of Apoptosis

Apoptosis

- ✓ is a form of 'coordinated and internally programmed cell death' in which cells activate enzymes that degrade the cells' own nuclear DNA and nuclear and cytoplasmic proteins.

- ✓ Apoptosis is a Greek word meaning ('falling off' or 'dropping off').

Causes of Apoptosis:

- Physiologic Apoptosis

- Pathologic Apoptosis

Physiologic Apoptosis

- **Death by apoptosis is a normal phenomenon that serves to eliminate cells that are no longer needed and to maintain a constant number of cells of various types in tissues.**

Apoptosis in Physiologic Situations:

- ✓ **Programmed destruction of cell during embryogenesis & development**
- ✓ **Involution of hormone-dependent tissues upon hormone deprivation, e.g**
  - **endometrial cells (menstrual cycle)**
- ✓ **Cell loss in proliferating cell populations,e.g. skin**

Apoptosis in Physiologic Situations:

- ✓ **Elimination of cells that have served their useful purpose, e.g**
  - **Post-inflammatory clean up (neutrophils).**
- ✓ **Cell death induced by cytotoxic T-cells to eliminate harmful cells, e.g**
  - **viral infected or tumor cells**
- ✓ **Elimination of potentially harmful self-reactive lymphocytes**

Apoptosis in Pathologic Conditions:

- **Apoptosis eliminates cells that are genetically altered or injured beyond repair and does so without eliciting a severe host reaction.**

Apoptosis in Pathologic Conditions:

* **DNA damage.** (Radiation, cytotoxic anticancer drugs)
* **Cell injury in certain infections.** (viral infections)
* **Pathologic atrophy in parenchymal organs after duct obstruction.** (pancreas, parotid gland)

Two main pathways:

- **Intrinsic 'mitochondrial' pathway:**

Increased permeability of mitochondrial membrane results in release of pro-apoptotic factors (cytochrome c) that activate downstream caspases →death .

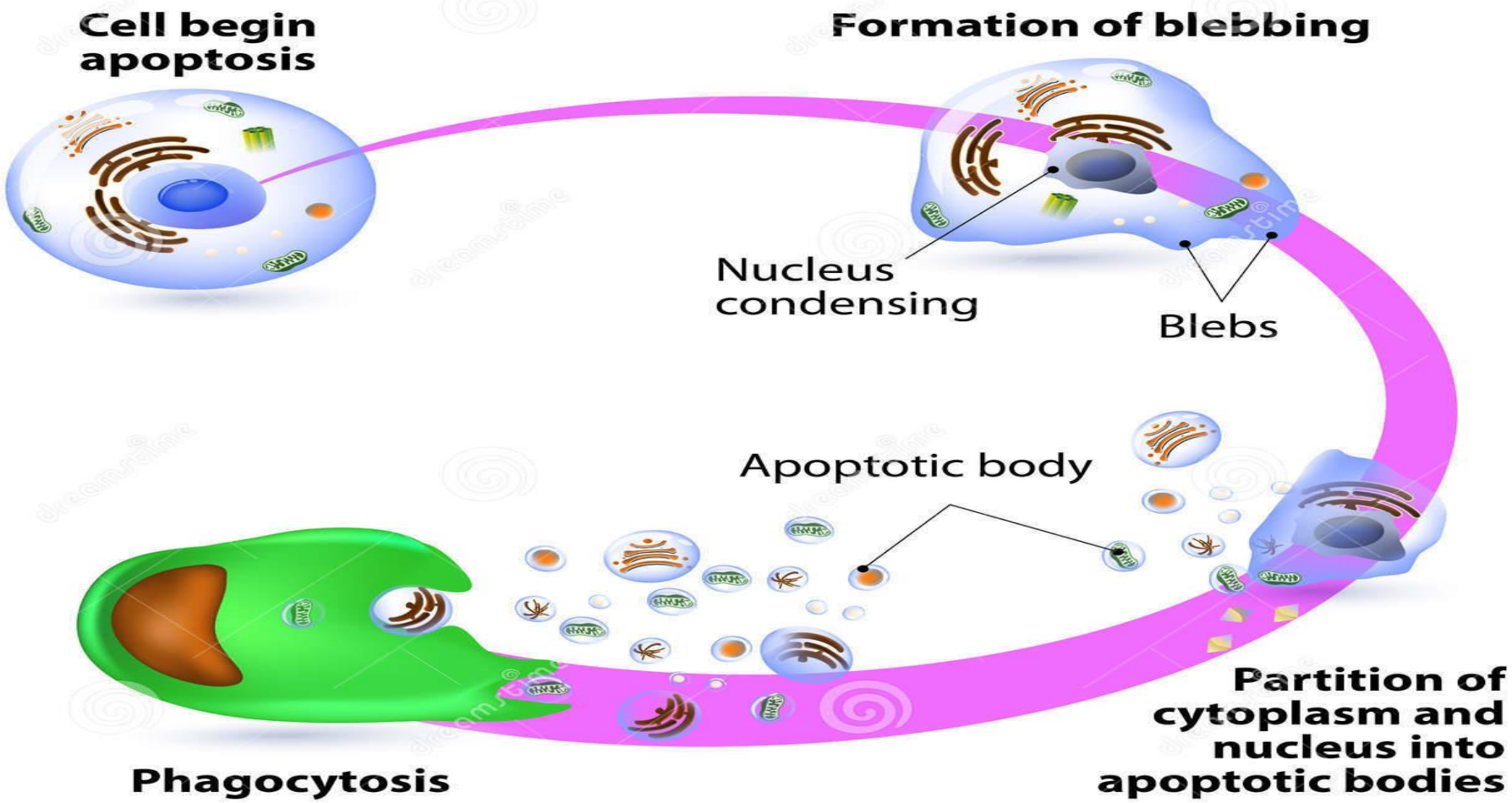
- **Extrinsic 'death receptor' pathway:**

FAS and TNF1 receptor families with death domain.

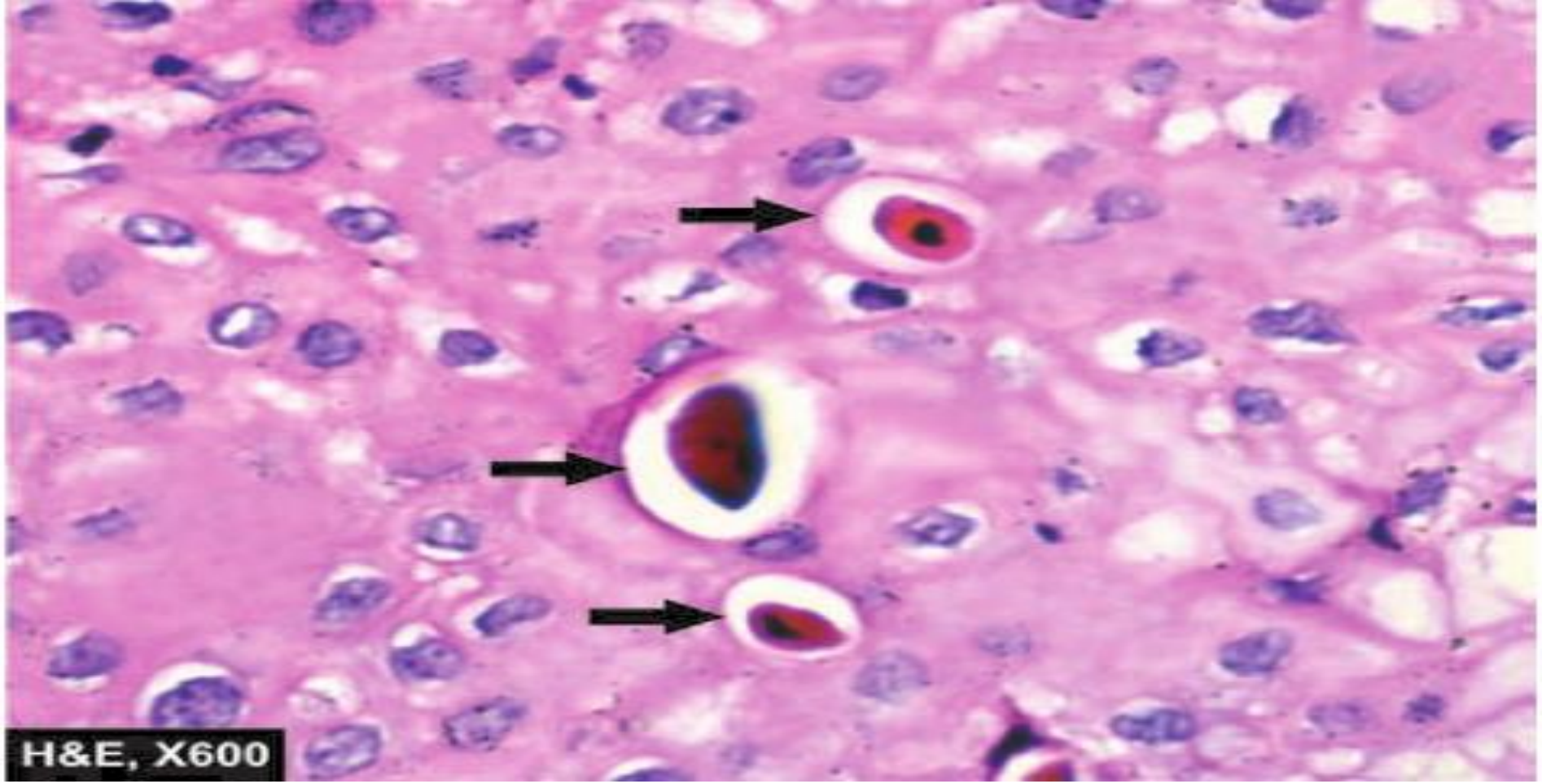
Morphology Of Apoptosis

- Involvement of single cells or small clusters of cells.
- Cell shrinkage.
- Chromatin condensation and fragmentation.
- Formation of cytoplasmic blebs and apoptotic bodies.
- Phagocytosis of apoptotic bodies by adjacent healthy cells or macrophages.
- Lack of inflammation.

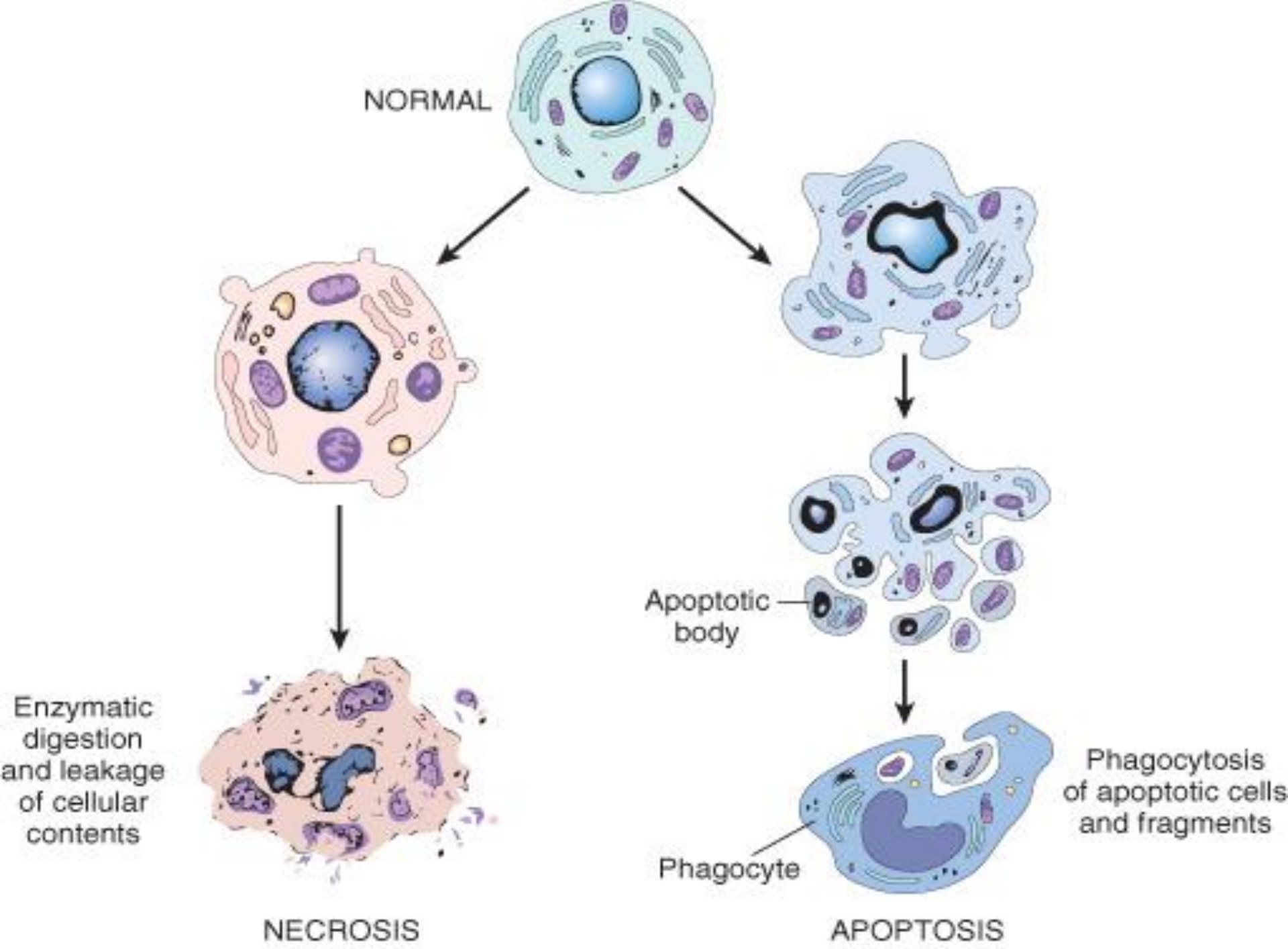
APOPTOSIS



Apoptosis consists of condensation of nuclear chromatin and fragmentation of the cell into membrane-bound apoptotic bodies which are engulfed by macrophages.



Apoptotic bodies in the layer of squamous mucosa. The dead cell seen in singles, is shrunken, the nucleus has clumped chromatin, while the cytoplasm in intensely eosinophilic.
There is no inflammation, unlike necrosis.


NORMAL
Enzymatic digestion and leakage of cellular contents
Apoptotic body
Phagocytosis of apoptotic cells and fragments
Phagocyte
NECROSIS
APOPTOSIS

Feature	Necrosis	Apoptosis
Cell size	Enlarged (swelling)	Reduced (shrinkage)
Nucleus	Pyknosis → karyorrhexis → karyolysis	Fragmentation into nucleosome-sized fragments
Plasma membrane	Disrupted	Intact; altered structure, especially orientation of lipids
Cellular contents	Enzymatic digestion; may leak out of cell	Intact; may be released in apoptotic bodies
Adjacent inflammation	Frequent	No
Physiologic or pathologic role	Invariably pathologic (culmination of irreversible cell injury)	Often physiologic means of eliminating unwanted cells; may be pathologic after some forms of cell injury, especially DNA and protein damage

Thank you